Readar: Where Books Spark Connections

Xi'an Jiaotong-Liverpool University 西交利物浦大学 CPT208 Human-Centric Computing **Group 42**

Group42 Members: *Junhao Huang 2256792*
Ruiyang Wu 2257475
Mengyao Jiang 2252837
Weizhe Sun 1927150
Daiyan Wu 2257387

01 Introduction

Domain:

Generative AI in Deep Reading & Chatting Assistance

> *"With the rise of Generative AI, HCI in social computing has advanced rapidly. One yet underexplored area is using AI to support deep reading and meaningful discussion."*

- Our project, **Readar** is an **AI-powered platform that connects readers for one-on-one, topic-focused discussions**. By analyzing reading behavior, annotations, and shared viewpoints, it matches users and **facilitates thoughtful, structured conversations**.
- Unlike casual chat apps, Readar offers **AI-generated icebreakers**, **real-time guidance and summaries** to **keep discussions insightful and on track**. It promotes **deep engagement with content**—turning reading into a richer, more interactive experience.

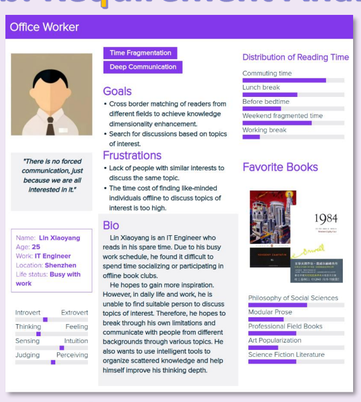
02 Background Statement

a. Requirement & Problem Discovery

Interview: A semi-structured many-to-one interview was conducted, including 2 undergraduates and 1 book club organizer. Focused on reading habits, annotation preferences, discussion experiences, and expectations from AI-assisted reading tools.

Survey: An online questionnaire [1] was distributed featuring both open-ended and closed-ended questions, covering reading behavior, annotation practices, discussion habits, and attitudes toward AI functionalities. The survey collected 37 valid responses. Besides, we also studied two related survey reports [2], [3].

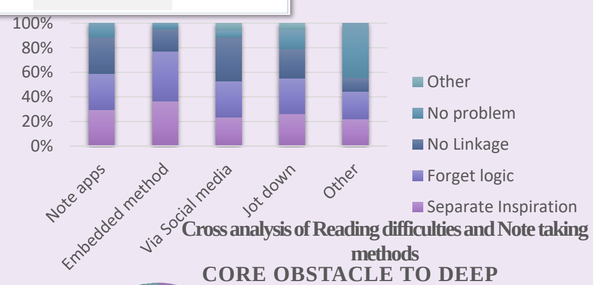
b. Requirement Analysis



↑Scan the code to view Detailed Survey Analysis [1]

Need of Integrated Insight Capture

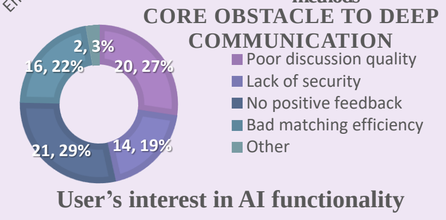
Most users **(59%)** record ideas informally, while little **(30%)** use in-app annotations. Though **62%** agree annotations aid understanding, only **24%** highlight.



Cross analysis of Reading difficulties and Note taking methods

Discussion Desire vs. Social Barriers

While **51%** wish to discuss resonant content, **84%** fail due to mismatched reading experiences.

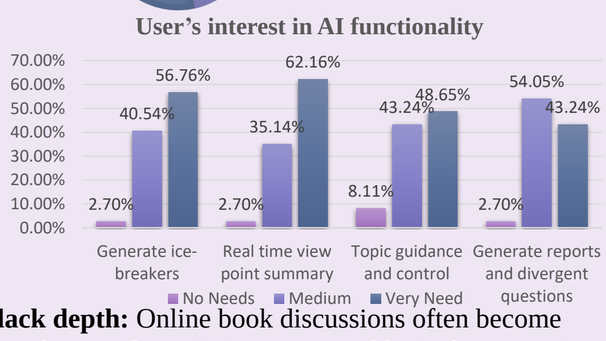


AI-Driven Conversation Support

Great interest in AI functionalities: Valuing AI features like icebreaker generation **(57%)** and in-text quote support **(76%)**.

Privacy and Data Sharing Considerations

Most users willing to share non-sensitive data: reading behavior **(81%)** , annotated content **(67.57%)** and anonymized chats **(78%)**.



c. Problems to Address

- **Discussions on public platforms lack depth:** Online book discussions often become shallow or redundant, full of emotional output but missing structured logical progression.
- **Social dynamics hinder honest opinions:** Group discussions tend to promote social conformity, discouraging controversial or unconventional viewpoints.
- **Finding a suitable discussion partner is difficult:** Reading same book not guarantee intellectual compatibility. Readers need macro-level idea alignment, not just shared book titles.
- **Initiating discussions is intimidating:** Many readers struggle with starting or sustaining deep discussions, especially where they might be misunderstood or challenged aggressively.

d. Key Difference from Existing Works

- **Traditional social platforms and reading communities** (e.g., Douban, Goodreads): discussions are mostly group interactions, easily lead to pandering and disengagement.
- **Personalized reading tools**: Some support annotation or sharing but lack targeted discussion matching and AI-guidance. Fail to promote a true exchange of opinions.
- **Generative AI dialogue system**: Focus on Q&A but does not focus on the multi-dimensional interaction and logical extension of in-depth reading and discussion.

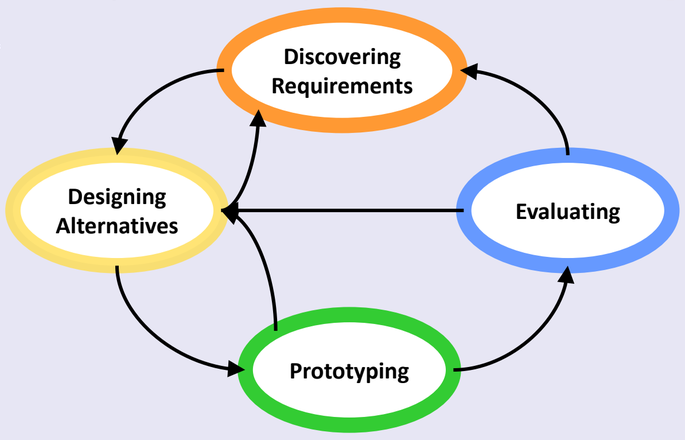
e. Importance & Relevance to Human-Centric Computing

Readar is designed to personalize and enhance human discussion experiences. By leveraging AI-driven profiling and conversation facilitation, our system ensures logical, pressure-free exchanges, preserving individual opinions while fostering deeper engagement.

03 Methodology & Implementation

- We began by **identifying the key problem** in digital reading: users struggle to engage in **meaningful post-reading discussions**. To validate this issue, we conducted **user research** by surveys [1] gather pain points and what deter people from joining discussion. Then, synthesized feedback to create **user personas** and **scenarios**, highlighting involving people's core demands to do data analysis. Based on the analysis to confirm the **feature prioritization**.

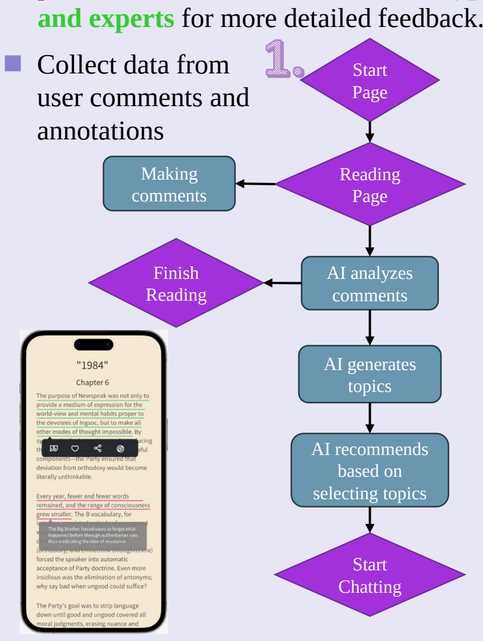
- In finding alternative phase, we **brainstormed multiple possible solutions** to handle the problem that we collected from last phase. We evaluated whether design criteria alignment with user needs and technical feasibility of human-centered generative AI.



- Through **three design iterations**, we refined "Readar" by doing **interviews** to collect feedback on usability, as well as **expert's reviews** to do heuristic evaluation to identify UX and UI changes. To ensure our improvement and the modification priority are data-driven.

- In this part, we translated ideas into tangible prototypes, and we mainly created a **flowchart** to map user's entire operation process and an **Interactive UI Prototypes** as wireframes and high-fidelity interfaces . We presented prototypes to **users and experts** for more detailed feedback.

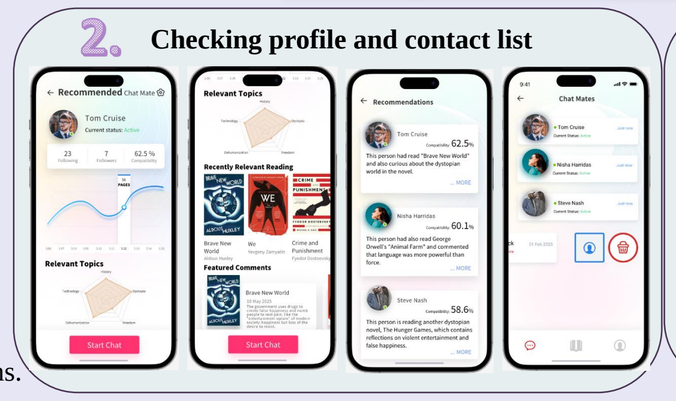
- Collect data from user comments and annotations



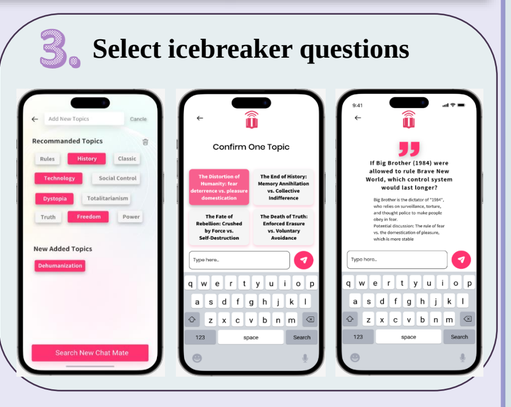
Function / Iteration	Key information collection	AI matching mechanism	AI-assisted conversation
Iteration I	Collect reader comments, and annotations; Record reader reading time; Record reader's mood fluctuation.	Search who recently read the same book and have written similar comments; With similar time distribution and mood swings.	AI generate icebreaker topics about overlapping viewpoints; Present the corresponding comment during conversation;
Iteration II	Remove the mood fluctuation record; Record the reader's annotations and viewpoints.	Expands search scope to books on the same subjects; AI focuses on matching viewpoints in comments.	AI summarizes the plot in the book read by another reader; AI generates real-time chat summaries and expansive topics.
Iteration III	AI records the key words that mentions multiple times in comments.	Search who shows interest in similar topics;	AI adds online search function; Show Notice board.

- Our group's prototype is designed and **iteratively improved** based on the user demands and suggestions that we gathered during the previous iterations.

2. Checking profile and contact list



3. Select icebreaker questions

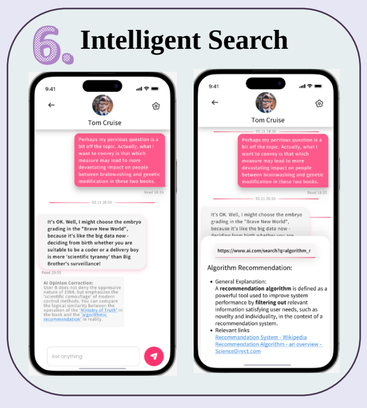


- To avoid excessive complex elements that could disturb user experience [3]. We try to keep the layout of the page as clear as possible and remove redundant elements which may cause confusions.

4. Avoid Discrimination & chat reports

5. Retrieve relevant comments

6. Intelligent Search



04 Results

Evaluation Method

To evaluate Readar's usability and user experience, we used two complementary methods:

- **System Usability Scale (SUS):** A ten-item questionnaire measuring ease of learning, operation and interface intuitiveness in real-world use.
- **Heuristic Evaluation**: Experts assessed design, interaction and accessibility based on usability principles to identify improvement areas.
- **A/B Testing**: Compared different designs to optimize usability and engagement.

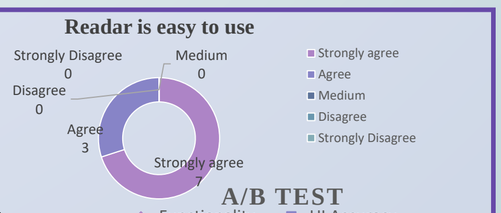
Feedback from 20 users, combined with **SUS scores**, **expert reviews** and **A/B testing**, offered a comprehensive understanding of Readar's strengths and areas for enhancement.

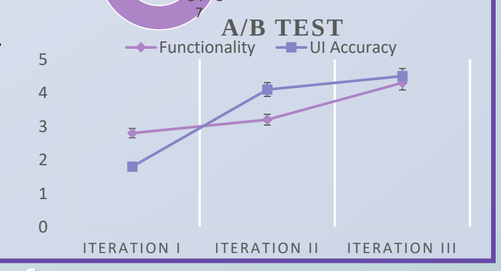
The heuristic evaluation identified key points:

- **Consistency**: Overall consistent, but some menu behaviors lack uniformity.
- **Feedback**: Clear in most cases, though some error messages need clearer guidance.
- **Efficiency**: AI features enhance engagement, but recommendation accuracy needs improvement.

Readar scored 75.6 on the SUS, well above the usability benchmark of 68 [1].

- 70% of users found it easy to use, citing its intuitive interface and low learning cost.
- 80% agreed features were clear and easy to understand.
- 95% reported few or no errors, reflecting strong stability and user-friendliness.
- **A/B testing**: Improved navigation boosted Readar's functionality and a new recommendation layout that meets user needs has increased intuitiveness.





Impact

Readar improves accessibility and user experience by reducing technical barriers. Users can generally master the core functions of Readar in a relatively short period of time, and users of different age groups have a high acceptance of Readar. Its intuitive interface and AI assistance effectively **reduce the time cost** of communication, **relieve pressure**, **promote deeper thinking** and **facilitate inclusive knowledge exchange**.

05 Conclusion

In Conclusion:

We developed Readar, a deep reading platform that helps users find suitable partners for discussion. By integrating generative AI during conversations, our app significantly enhances communication and reading experiences.

Limitations and Possible Solutions:

- The asynchronous communication model may reduce user engagement.
- GAI-powered conversations could lead to over-reliance on AI assistance.

To mitigate these problems, we propose an anti-reliance mechanism: When detecting user overly rely on AI assistance, the system will nudge users to engage in unaided discussions, fostering more organic interactions.

Reference

[1] R. Wu, 'Findings and Analyses from the Readar Questionnaire - CPT208G42', Xi'an Jiaotong-Liverpool University, Suzhou, China, 2025. [Online]. Available: https://a5eanegz59.feishu.cn/docx/KT0XdadeOoQn99xl35dc4uM2nSf

[2] iResearch Inc, '2020年中国数字阅读产品营销洞察报告-艾瑞数智'. Accessed: Apr. 06, 2025. [Online]. Available: https://www.idigital.com.cn/report/detail?id=3664

[3] Cheers Publishing, '《2024职场人阅读报告》 [in Chinese]'. Accessed: Apr. 06, 2025. [Online]. Available: https://mp.weixin.qq.com/s/UfMUC0-HnQmGLLYGre2raA